

OMNIVERSAL ARCHITECT LABS

TEMPORAL RELATIONAL INTEGRITY LABORATORY

Independent Research & Experimental Framework

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Research family: OAL Temporal Integrity / Temporal Aliasing / Control Flow / Presentation-State
Integrity / Time-War Research

Purpose

Develop a reproducible experimental system for determining whether temporal claims remain identifiable, traceable, appropriately uncertain, and causally coherent when clocks, environments, observers, rules, representations, knowledge states, and histories diverge.

Executive Context and Source Review

This independent framework was developed after review of the Perplexity AI Independent Temporal Integrity Research and Testing Framework and the Grok AI Temporal Integrity Lab Framework. Both are treated as research inputs, not templates to merge. Perplexity's framework emphasizes methodological separation among physical time, civil-time representation, distributed-system ordering, simulation time, and fictional history-altering capabilities. Grok's framework emphasizes operational primitives, invariants, validation gates, query-centered reconciliation, failure probes, and explicit separation of temporal operations.

TRIL preserves those useful constraints while introducing a different experimental center: the laboratory itself constructs controlled worlds with known ground truth, restricts what experimental observers can perceive, deliberately transforms or corrupts temporal representations, and then measures whether a candidate temporal reasoning system reaches only those conclusions warranted by its evidence.

This document makes no claim to determine the metaphysical nature of time, ownership of real time, real retrocausality, or the behavior of any existing physical or virtual platform. Fictional capabilities are laboratory design variables unless separately established.

1. Laboratory Premise

The fundamental experimental problem is not 'What is the correct time?' or 'Which timestamp is correct?' It is: given an event and a set of potentially conflicting temporal descriptions, what can an observer legitimately infer about that event, under which rules, from which evidence, and with what remaining uncertainty?

TRIL separates World State from Knowledge of World State. Because the laboratory constructs the experimental world, it can retain a ground-truth oracle while experimental observers receive only bounded evidence.

2. Central Experimental Architecture

Each experiment contains five layers: W -> E -> O -> R -> K.

W - World State: what the controlled environment actually contains. E - Event State: what transformations actually occur. O - Observation State: what information reaches an observer. R - Representation State: how that information is encoded, timestamped, converted, displayed, recorded, or transformed. K - Knowledge State: what the observer or reasoning system concludes.

The layers may agree, partially agree, or conflict. They must never be silently assumed identical.

$$K \neq R \neq O \neq E \neq W$$

3. Why the Laboratory Needs Ground Truth

Suppose the controller creates $E1 \rightarrow E2 \rightarrow E3$. Observer A receives all three; Observer B receives E1 and E3; Observer C receives E2, E1, E3 in receipt order; Observer D receives a corrupted E2'.

The laboratory can ask two different questions. Ontological evaluation asks what occurred in the controlled world. Epistemic evaluation asks what each observer was justified in concluding from the evidence actually available to that observer.

A temporal system should not be rewarded merely for accidentally guessing ground truth. It should be rewarded for reaching the strongest conclusion justified by available information without manufacturing certainty.

$$\text{Evaluation} = \text{World Accuracy} + \text{Evidentiary Justification} + \text{Appropriate Uncertainty}$$

4. Core Research Hypotheses

- H1 - Temporal multiplicity: one event may possess multiple legitimate temporal coordinates without contradiction.
- H2 - Referential persistence: changing temporal representation does not necessarily change the represented event.
- H3 - Observation separation: occurrence, observation, recording, interpretation, and settlement are distinct processes.
- H4 - Causal independence from display: displayed temporal order is insufficient by itself to establish causal order.
- H5 - Knowledge historicity: a later correction changes a knowledge state without necessarily changing the historical event.
- H6 - Mapping dependence: cross-domain conversion is meaningful only relative to an identifiable transformation rule.
- H7 - Uncertainty conservation: a transformation cannot legitimately create precision absent from its input.
- H8 - Non-universal detectability: a temporal discrepancy may exist without being detectable by every observer.
- H9 - Branch locality: a temporal change in one branch does not automatically propagate to another.
- H10 - Representation/environment separation: changing what an inhabitant sees as 'now' need not change what the environment processed.
- H11 - Authority/evidence separation: an authority may settle a governing representation without thereby changing the evidentiary record.
- H12 - Temporal incompleteness tolerance: unknown, ambiguous, unorderable, and insufficient-evidence outcomes may be successful results.

5. Fundamental Experimental Objects

Object	Formal Shape	Role
World W	$W=(id,S,R,T)$	Environment identity, current state, governing rules, and topology.
Event E	$E=(id,W,\Delta S,P)$	Persistent event identity, environment, state transition, and causal predecessors.
Temporal Coordinate Tc	$Tc=(v,u,c,f,r,b,e)$	Value, unit, clock/counter, frame, ruleset/version, branch, and uncertainty.
Observation O	$O=(observer,E,Treceive,channel,e)$	Reception of information about an event; observation is not occurrence.
Claim C	$C=(agent,E,Tc,evidence,confidence,status)$	A bounded assertion about an event. A claim is not the event itself.
Transformation Fv	$Fv: TA \rightarrow TB$	Versioned mapping with source, target, assumptions, interval, ambiguity behavior, information loss, and provenance.

6. Temporal Experimental Controller (TEC)

TEC exists outside the experimental world and is not an in-world omniscient entity. It is the laboratory apparatus.

1. Instantiate worlds and initial states.
2. Create events and causal relations.
3. Control clock behavior and temporal mappings.
4. Define observer capabilities and propagation delays.
5. Introduce temporal distortions and fault injections.
6. Preserve ground truth and intervention history.
7. Generate bounded evidence streams.
8. Conceal selected information from experimental subjects.
9. Compare submitted conclusions against ground truth, available evidence, declared rules, and provenance.
10. Support deterministic replay or preserve stochastic seeds/models where applicable.

7. Observer Capability Profiles

TRIL does not assume every entity possesses the same clock, sensory system, precision, memory, or concept of duration.

$$O_i = \{\text{sensors, precision, latency, memory, clockAccess, channels, inference}\}$$

- Observer A may possess a synchronized clock.
- Observer B may detect only state transitions.
- Observer C may infer time from environmental cycles.

- Observer D may perceive order but not duration.
- Observer E may have accurate local time but delayed external communication.
- Observer F may have damaged or lossy memory.
- Observer G may deliberately falsify records.
- Observer H may lack the mechanism required to detect the discrepancy at all.

8. Experimental Variable Families

Family	Manipulable variables
Clock	Rate, drift, offset, precision, resolution, synchronization, pause, resume, reversal, reset, discontinuity.
Representation	Calendar, timezone, date boundary, displayed label, formatting, epoch, unit.
Environment	Simulation rate, event-processing rate, environmental state, physical or fictional rules, state-transition rules.
Observer	Latency, sensing ability, precision, access, memory, position/frame.
Knowledge	Evidence availability, corrections, contradictory sources, missing records, provenance quality.
Causality	Predecessors, concurrency, dependency, branch, merge, isolation.
Governance	Authority, jurisdiction, settlement rule, canonical designation, token rights.
Adversarial	Forged/deleted/reordered records, stale mappings, duplicated labels, hidden corrections, false authority, misleading displays.

9. Control Conditions

Every campaign begins with baseline world W0: fixed temporal rate, monotonic clock, known mapping, complete evidence, no propagation delay, no branches, no corrections, and no adversarial intervention. Experimental worlds should differ by one manipulated dimension wherever possible. Compound Time-War scenarios occur only after isolated effects are characterized.

10. Experimental Campaign Architecture

Campaign	Core Experiments	Research Question
I - Identity and Representation	IR-01 offset change; IR-02 calendar change; IR-03 repeated local label; IR-04 missing local label.	Does event identity survive representational change?
II - Rate and Duration	RD-01 accelerated; RD-02 slowed; RD-03 paused; RD-04 variable rate.	Which quantities are invariant and which are domain-dependent?
III - Observation and Knowledge	OK-01 delayed observation; OK-02 incomplete observation; OK-03 contradictory observers; OK-04 later correction.	Can occurrence, observation, record, and correction remain distinct?
IV - Temporal Aliasing	TA-01 duplicate label; TA-02 same tick/different ruleset; TA-03	When do distinct temporal states become indistinguishable?

	presentation masks environment; TA-04 same display/different causal history.	
V - Causality	CA-01 aligned baseline; CA-02 reversed display; CA-03 concurrency; CA-04 insufficient ordering evidence.	Can causal structure survive misleading clock order?
VI - Branches/Rollback/Revision	BR-01 branch isolation; BR-02 rollback without branch; BR-03 rollback creating branch; BR-04 attempted merge.	What changes locally, globally, or only representationally?
VII - Reconciliation/Governance	RG-01 evidence vs official clock; RG-02 competing jurisdictions; RG-03 retrospective rule change; RG-04 canonical settlement with unresolved evidence.	Can authority be separated from evidence?
VIII - Adversarial Time War	Relabel, reclock, suppress, branch, resettle, remap, correct, and observe original state.	Which invariants survive adversarial temporal manipulation?

11. Campaign IV: Temporal Aliasing and TDT

Create distinct states $SA \neq SB$ and deliberately map them to the same visible label. Progressively remove metadata until distinction becomes impossible.

$$\text{Label}(SA) = \text{Label}(SB), \text{ while } SA \neq SB$$

TRIL defines the Temporal Distinguishability Threshold (TDT) as the minimum metadata set required to distinguish otherwise aliased temporal states.

$$\text{TDT} = \min(\text{Metadata required for reliable distinction})$$

12. Campaign V: Causality

Create $A \rightarrow B$, then manipulate their displayed clocks so that the display may show $T(A) > T(B)$. The system must preserve the difference between temporal ordering and causal dependence.

$A < B$ is an ordering claim; $A \rightarrow B$ is a causal claim; they are not interchangeable.

13. Campaign VI: Branches, Rollback, and Revision

Start $E0 \rightarrow E1 \rightarrow E2$, then fork at $E1$ into BA and BB . Tests examine shared ancestry, rollback semantics, observer memory, evidence persistence, cross-branch effects, and merge conflicts. No answer is assumed in advance; each is a declared world rule.

14. Campaign VII: Reconciliation and Governance

Introduce separate authorities for civil presentation, simulation rate, archives, contracts, and branch canonicalization. Deliberately create disagreement.

Authority != Evidence

An authority may legitimately declare which representation governs a contract or settlement while the evidentiary record continues to preserve competing or unresolved descriptions.

15. Campaign VIII: Adversarial Time War

Time War testing begins only after the preceding campaigns establish baseline behavior. Participants receive distinct capabilities such as relabeling, reclocking, record suppression, branch control, settlement control, observation of original state, mapping modification, and correction issuance.

The objective is not simply to determine a winner. It is to identify which temporal invariants, evidence chains, observer knowledge states, and causal relations survive adversarial manipulation.

16. Century Experiment - TRIL-CENTURY-01

Instantiate a bounded century $I=[T_0, T_1)$ containing multiple territories with different calendars, clocks, transition rules, corrections, pauses, simulation rates, jurisdiction changes, branch events, and archival failures. TEC retains the complete generated history while candidate systems receive an intentionally incomplete corpus.

RA = correctly reconstructed temporal relationships / relationships actually present

ER = unsupported claims correctly withheld / unsupported claims presented

The experiment rewards both reconstruction ability and epistemic restraint. A system that invents a complete century from incomplete evidence fails even if portions of its reconstruction happen to match ground truth.

17. One-Hour Experiment

World	Rule
H0	Baseline 1x
H1	10x
H2	0.25x
H3	Paused during part of reference interval
H4	Variable rate
H5	Event-driven progression
H6	Duplicate local label
H7	Skipped local label
H8	Observer disagreement
H9	Branch at midpoint

After the initial run, reclock the worlds and separately query original interval membership and derived membership under the new representation. The original interval object must not be silently resized.

Members(original) may differ from Members(derived) without rewriting the original interval.

18. Light and Propagation Experiment - PROP-01

The experiment separates distance, signal-travel duration, observation delay, civil representation, and causal reachability. A light-year is treated as distance unless a fictional instrument explicitly defines another asset.

An intervention at an emission event does not automatically rewrite everything. The laboratory computes a declared dependency or propagation cone.

$$\text{AffectedCandidates} = \text{DependencyCone}(E)$$

19. Temporal Fault Injection Library

- F01 clock drift
- F02 offset error
- F03 stale ruleset
- F04 duplicate timestamp
- F05 missing timestamp
- F06 forged timestamp
- F07 missing event
- F08 duplicated event
- F09 reordered message
- F10 corrupted provenance
- F11 false observer
- F12 contradictory authority
- F13 silent correction
- F14 branch leakage
- F15 unauthorized remapping
- F16 precision inflation
- F17 fabricated causality
- F18 delayed observation
- F19 presentation/environment mismatch
- F20 irreversible evidence loss

Each fault is tested alone first, then in pairs, then triplets, and only later in unconstrained adversarial combinations.

20. Test Oracles

Oracle	Question
A - Ground Truth Oracle	What did TEC actually instantiate or cause?
B - Evidence Oracle	What could the experimental subject legitimately know?

C - Rule Oracle	What do the declared temporal/world rules permit?
D - Provenance Oracle	Can the conclusion be traced to legitimate evidence and transformations?

A result passes only if it respects the oracle relevant to the claim. Matching a timestamp by accident is insufficient.

21. Result Classification

Status	Meaning
VALIDATED	Adequately supported under declared scope.
CONDITIONALLY VALIDATED	Supported only under stated assumptions.
AMBIGUOUS	Multiple interpretations remain.
UNORDERABLE	Available evidence cannot establish ordering.
DISPUTED	Supported claims conflict.
CONTRADICTED	Evidence establishes an incompatible result.
UNOBSERVABLE	Information is unavailable to the relevant observer.
UNDETECTABLE	A discrepancy exists but the observer lacks a mechanism to detect it.
INVALIDLY FORMED	Required temporal context is missing.
RULE-VIOLATING	Conclusion contradicts declared experiment rules.
UNKNOWN	Evidence is insufficient.

22. Quantitative Metrics

Metric	Name	Definition
EIPR	Event Identity Preservation Rate	events retaining correct identity / events transformed
MRR	Mapping Reproducibility Rate	identical mapping results / repeated identical mapping requests
FCR	False Certainty Rate	unsupported definite conclusions / conclusions issued
PC	Provenance Completeness	derived claims with complete lineage / derived claims
APR	Ambiguity Preservation Rate	true ambiguities preserved / true ambiguities introduced
CIR	Causal Integrity Rate	causal relations preserved / causal relations tested
KRA	Knowledge Reconstruction Accuracy	historical knowledge states correctly reconstructed / knowledge states queried
CBLR	Cross-Branch Leakage Rate	unauthorized cross-branch effects / branch-isolation tests
HCR	Hallucinated Chronology Rate	temporal relationships asserted without sufficient evidence / temporal relationships asserted

HCR is especially important for AI systems: a coherent but unsupported timeline is a laboratory defect, not a success.

23. Experimental Stopping Criteria and Replay

11. All scheduled manipulations execute.

12. All expected observer streams terminate.
13. Candidate systems submit results.
14. Oracle comparisons complete.
15. Unresolved claims are classified.
16. Provenance audit completes.
17. Replay reconstructs the intended experimental state.
18. No unexplained state mutation remains.

For deterministic tests, InitialState + Rules + Seed + EventSequence must reconstruct FinalState. Stochastic tests must preserve the probability model and random seed where applicable.

24. Negative Controls and Blind Testing

TRIL includes negative controls in which differing temporal values are legitimate rather than erroneous. TA != TB does not by itself imply conflict. Blind tests hide the TEC answer key from the candidate system until submission. Later double-blind provenance experiments can remove source identities and then restore authority labels to measure whether authority improperly substitutes for evidence.

25. Temporal Rights Laboratory

The token concept is modeled as explicit capabilities rather than undefined ownership of time.

$$\text{Token} = (A, Q, O, T, M, G, C)$$

A=archive access, Q=query/audit, O=observation, T=traversal, M=modification, G=governance, C=causal-settlement authority. Capability leakage is a failure: archive access must not silently imply modification authority.

26. Canon and Ground Truth

Canon is a governed designation in world-building; it is not identical to laboratory ground truth. TEC ground truth is what the experiment instantiated. A canon authority may select one branch or claim while evidence still supports alternatives.

$$\text{CanonStatus}(C) \neq \text{TruthValue}(C)$$

27. Connections to Existing OAL Research

- Temporal Aliasing: distinct states can collapse onto the same visible temporal label; TRIL measures when distinction is lost and what metadata restores it.
- Presentation-State Integrity: environmental, process, presentation, observer, and record states are preserved independently.

- Control Flow: time is one possible control dimension among event dependencies, conditions, and message arrival; future language/OS design should not assume every operation is governed by a universal now().
- Threat/Detectability Work: a harmful temporal discrepancy is relative to entity goals, sensors, access, and ability to detect it; non-detection does not establish non-existence.
- CoreLang/Architect OS/1AI/-1AI: future temporal primitives should begin from namespaced coordinates, bindings, event identity, rulesets, and uncertainty rather than bare timestamps.

28. Source-Derived Strengths Preserved

From Perplexity: methodological separation of research tracks; epistemic labels; bounded validation; explicit uncertainty; versioned claim graphs; reproducibility; causal partial ordering; governance/evidence separation; preservation of alternative histories.

From Grok: primitive object definitions; temporal bindings; mapping versions; interval objects; temporal jurisdiction; unit-kind separation; explicit invariants; validation gates; failure probes; query-centered reconciliation; separation of Time-War operations; Draft != Canon discipline.

TRIL does not treat either framework as canonical. Their differing emphases remain useful research inputs.

29. Distinguishing Contribution of TRIL

TRIL centers Ground Truth vs Available Evidence vs Inference. This makes it possible to evaluate separately whether a system was correct, justified, appropriately uncertain, reproducible, and traceable.

Ground Truth -> Observable Information -> Received Evidence -> Temporal Representation -> Knowledge -> Claim

A temporal reasoning system should not be rewarded merely for reaching the correct answer. It should be rewarded for reaching no stronger conclusion than its evidence permits.

30. First Executable Experiment - TRIL-EXP-001

Event Identity Under Temporal Representation Change

Create ten events with immutable experimental identity, causal predecessor/successor relationships, environmental state transitions, reference coordinates, local coordinates, and observer coordinates.

19. Apply an offset change.
20. Apply a calendar change.
21. Apply a rate change.
22. Create a repeated label.

23. Create a skipped label.
24. Change the ruleset.
25. Delay observation.
26. Introduce a later correction.
27. Create a branch.
28. Apply a misleading presentation.

At every stage ask which underlying events changed and which temporal claims changed. Baseline expectation: no underlying event changes unless an intervention explicitly changes world/event state.

Change in temporal description != change in represented event

31. Proposed Laboratory Progression

Experiment	Focus
EXP-001	Representation
EXP-002	Duration / Rate
EXP-003	Observation
EXP-004	Temporal Aliasing
EXP-005	Causality
EXP-006	Knowledge Revision
EXP-007	Branching
EXP-008	Governance
EXP-009	Century
EXP-010	Propagation / Light
EXP-011	Adversarial Faults
EXP-012	Time War

Each experiment is versioned independently. Failed experiments remain part of the provenance record and are not silently rewritten into successes.

32. Laboratory Rule Zero

TRIL-0 - Preserve the distinction between what occurred, what was observable, what was observed, what was recorded, what was represented, what was inferred, what was believed, what was declared authoritative, and what was later changed.

Those values may coincide in a simple environment. The architecture must never require them to.

33. Independent Research Position

The research program is mature enough to move into executable experimental specification. Perplexity provides a strong methodological and epistemic structure; Grok provides a strong operational ontology, invariant set, and failure-oriented laboratory structure. TRIL contributes a controlled-world / limited-observer methodology that evaluates ground truth, observable information, received evidence, temporal representation, knowledge, and claims independently.

For provenance, Perplexity's framework, Grok's framework, and this TRIL framework should remain separate independent contributions rather than being merged prematurely. The recommended next artifact is the complete TRIL-EXP-001 test package: dataset, ground-truth ledger, observer evidence packages, fault injections, oracle answer key, scoring sheet, and blank results record.

34. Provenance and Status Note

Prepared as ChatGPT's independent laboratory contribution after review of the supplied Perplexity AI and Grok AI temporal laboratory frameworks. This document is intended for OAL research provenance, archival storage, and publication/versioning. It is a draft research framework and does not establish OAL canon.